

RESEARCH PROGRESS BRIEF

Cross-dataset anticancer drug-response modeling

Gene-level versus Hallmark pathway representations across one exploratory pilot and a prespecified five-drug extension

Franbon Ahmed Mohammed

MS in Business Analytics Candidate, George Washington University

Status: preliminary, non-peer-reviewed cell-line research

Executive summary

- Trametinib was first analyzed as an exploratory proof of concept. A five-drug panel—gemcitabine, docetaxel, talazoparib, vorinostat, and bortezomib—was then frozen before confirmatory analyses.
- Models were trained on PRISM drug response plus DepMap baseline expression and evaluated on held-out GDSC cell lines. No GDSC response outcome was used for training, preprocessing, feature selection, or tuning.
- The prespecified primary comparison used Elastic Net, LN_IC50, strict GDSC2 Sanger-expression cohorts, and paired-bootstrap confidence intervals.
- In GDSC2, pathways were significantly favored for three drugs, individual genes for one, and one comparison was uncertain. GDSC1 did not reproduce the pathway advantage.

Main conclusion

Across the prespecified drug panel, whether genes or pathways performed better depended on the drug, model, and screening dataset. The inconsistent GDSC2–GDSC1 direction indicates that dataset shift is central to the scientific problem and that no representation should be declared universally superior.

Completed component	Verified scope
Exploratory pilot	Trametinib; internal validation, GDSC2 test, GDSC1 replication
Prespecified extension	Five drugs representing five mechanism classes
Representations	1,000 selected genes versus 50 Hallmark pathway scores
Models	Elastic Net and Random Forest, with a mean baseline
Uncertainty	1,000 paired bootstrap resamples per external comparison

1. Scientific question and study design

Question: Does compressing baseline gene expression into biologically defined Hallmark pathway scores improve cross-dataset prediction of anticancer drug response relative to selected individual genes?

Rationale: Pathways may reduce dimensionality and improve biological interpretability, but averaging can also erase drug-specific signals. External validation is required because internal cross-validation alone does not measure transfer across laboratories, expression pipelines, and drug screens.

Data flow

Stage	Input and key	Output / analytical role
Training response	PRISM; DepMap model ID	One drug-response target per training cell line
Training features	DepMap expression; ModelID	Baseline log-transformed gene expression
Identity control	DepMap Model.csv; ACH ↔ SIDM	Exclude mapped PRISM identities from GDSC tests
External response	GDSC1/GDSC2; SANGER_MODEL_ID	LN_IC50 primary; AUC secondary
External features	Sanger RNA-seq; SANGER_MODEL_ID	Strict external predictor matrix
Feature alignment	Cleaned gene_symbol	19,204 common genes in every drug analysis

Modeling protocol

- One row represented one cancer cell line. Predictors X were baseline expression values or derived pathway scores; response y was the PRISM drug-response value.
- Gene models used median imputation, univariate training-only selection of 1,000 genes, and Elastic Net scaling/regularization or a 300-tree Random Forest.
- Pathway models converted 4,374 matched genes into 50 Hallmark scores using normalization learned from PRISM training data only.
- Five-fold internal cross-validation assessed in-dataset performance. Final models were refit on all PRISM training rows and used once to predict held-out GDSC cohorts.
- External agreement was measured with Pearson and Spearman correlations. Paired pathway-minus-gene differences used 1,000 bootstrap resamples.

Sample sizes and quality-controlled dimensions

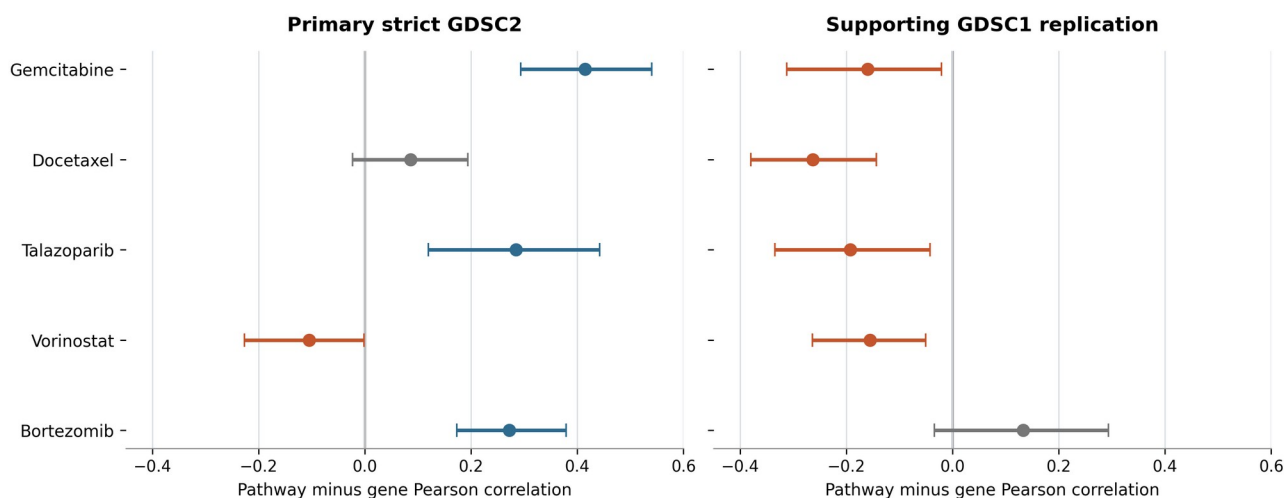
Drug	Role	PRISM train	GDSC2 strict	GDSC1 strict
Trametinib	Pilot	551	351	319
Gemcitabine	Extension	532	349	335
Docetaxel	Extension	556	351	336
Talazoparib	Extension	553	347	323
Vorinostat	Extension	522	352	339
Bortezomib	Extension	552	348	191

2. Prespecified primary results

Primary estimand: Elastic Net pathway-minus-gene Pearson correlation for LN_IC50 in strict held-out GDSC2 cell lines. Positive values favor pathways; a 95% interval excluding zero determines the favored representation.

Drug	n	Gene r	Pathway r	Difference [95% CI]	Conclusion
Gemcitabine	349	-0.194	+0.221	+0.414 [+0.294, +0.540]	Pathways favored
Docetaxel	351	+0.134	+0.220	+0.086 [-0.023, +0.194]	difference uncertain
Talazoparib	347	+0.086	+0.371	+0.284 [+0.120, +0.442]	Pathways favored
Vorinostat	352	+0.111	+0.006	-0.105 [-0.226, -0.002]	Genes favored
Bortezomib	348	-0.112	+0.160	+0.272 [+0.173, +0.379]	Pathways favored

Prespecified Elastic Net gene-versus-pathway comparison



Positive values favor Hallmark pathways. Error bars are paired-bootstrap 95% confidence intervals.

Interpretation

- Pathways were significantly favored for gemcitabine, talazoparib, and bortezomib.
- Genes were significantly favored for vorinostat. The docetaxel interval crossed zero.
- A representation difference does not by itself establish strong absolute prediction. For example, vorinostat Elastic Net correlations were weak for both representations.

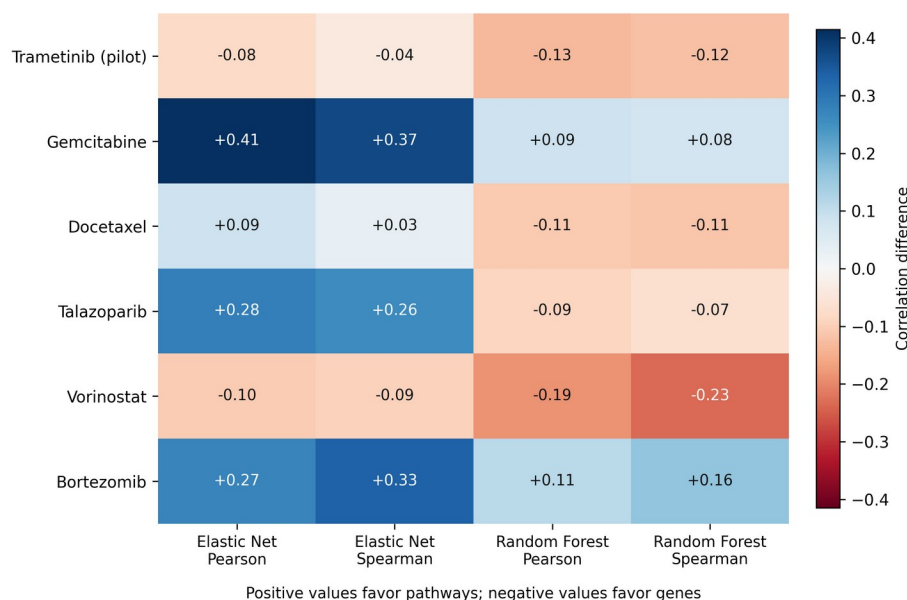
These results reject a simple universal rule: pathway aggregation can help for some drug–dataset combinations, but it can also reduce or fail to improve transfer performance.

3. Replication, model dependence, and the pilot

GDSC1 replication: The same prespecified Elastic Net comparison shifted in the opposite direction for four drugs; those four confidence intervals favored genes. Bortezomib was uncertain. GDSC1 and GDSC2 share many cell-line identities, so this is a cross-screen replication rather than a fully independent biological cohort.

Drug	n	Gene r	Pathway r	Difference [95% CI]	Conclusion
Gemcitabine	335	+0.150	-0.010	-0.160 [-0.312, -0.021]	Genes favored
Docetaxel	336	+0.354	+0.091	-0.263 [-0.380, -0.144]	Genes favored
Talazoparib	323	+0.228	+0.035	-0.193 [-0.335, -0.043]	Genes favored
Vorinostat	339	+0.091	-0.064	-0.155 [-0.264, -0.051]	Genes favored
Bortezomib	191	+0.109	+0.242	+0.133 [-0.034, +0.293]	difference uncertain

GDSC2 strict pathway-minus-gene differences



Cross-analysis summary

- Across the 20 GDSC2 extension comparisons (five drugs × two models × two correlations), eight favored pathways, six favored genes, and six were uncertain.
- Across the corresponding 20 GDSC1 comparisons, 12 favored genes, eight were uncertain, and none significantly favored pathways.
- The exploratory trametinib pilot favored genes or was uncertain in GDSC2; it is displayed for context but excluded from confirmatory inference.

4. Quality controls, limitations, and next steps

Leakage and reproducibility checks

- All six analyses aligned 19,204 common gene symbols, selected 1,000 genes inside PRISM training, and constructed 50 pathways from 4,374 matched member genes.
- The external-feature audits record zero GDSC outcomes used during training for every gene and pathway model.
- Strict GDSC2 cohorts contained 347–352 held-out Sanger-expression models per extension drug; strict GDSC1 cohorts contained 191–339.
- All five extension drugs completed without missing outputs and used the same frozen panel configuration. When multiple GDSC screens existed, selection was performance-blind.

Principal limitations

Limitation	Consequence
Cell-line experiments	Results do not establish patient benefit or clinical utility.
Different response scales	PRISM response and GDSC LN_IC50 are compared by correlation, not directly calibrated error.
Cross-platform dataset shift	Differences may reflect laboratory, screen, expression, and cohort effects.
Simple pathway scoring	Mean standardized expression may discard directional or context-specific biology.
Small drug panel	Five confirmatory drugs cannot support claims across all anticancer therapies.
GDSC overlap	GDSC1 is supporting cross-screen evidence, not a completely independent cell-line cohort.

Recommended next stage

- Freeze the current code, tables, figures, and prespecified results as version 1.0; do not tune models after seeing these outcomes.
- Seek faculty review of the design, especially cross-screen heterogeneity, response-scale harmonization, and pathway scoring.
- Add drug-level and dataset-level heterogeneity analyses only under a written secondary-analysis plan.
- Prepare a poster abstract after methodological review. Consider a preprint only after the repository is independently reproduced and the manuscript receives scientific supervision.

References

- Corsello SM et al. Discovering the anticancer potential of non-oncology drugs by systematic viability profiling. *Nature Cancer*. 2020. doi:10.1038/s43018-019-0018-6.
- Ghandi M et al. Next-generation characterization of the Cancer Cell Line Encyclopedia. *Nature*. 2019. doi:10.1038/s41586-019-1186-3.
- Iorio F et al. A landscape of pharmacogenomic interactions in cancer. *Cell*. 2016. doi:10.1016/j.cell.2016.06.017.
- Liberzon A et al. The Molecular Signatures Database Hallmark gene set collection. *Cell Systems*. 2015. doi:10.1016/j.cels.2015.12.004.
- Zou H, Hastie T. Regularization and variable selection via the elastic net. *JRSS B*. 2005. Breiman L. Random Forests. *Machine Learning*. 2001.

Research boundary: This report describes reproducible computational analyses of public cancer cell-line datasets. It does not provide clinical recommendations, causal biomarker claims, or evidence of patient-level effectiveness.